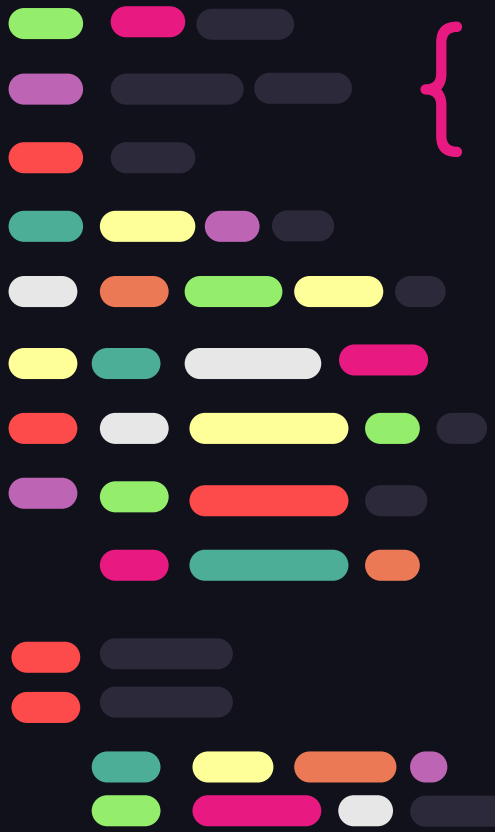
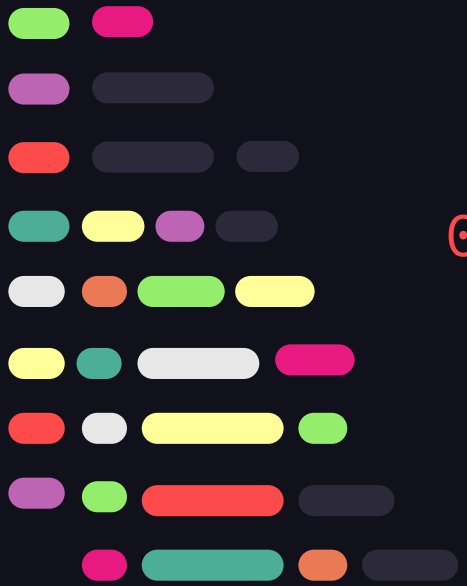


Introduction to Object Oriented Programming

CSE 108 January 2024

Implementation of Stack



01 Classical Approach

02 Encapsulation of Variables

03 Encapsulation of Functions

04 Full OOP with Class

Stack Implementation in C

Structured Programming Approach

- Global variables: Stack (`char s[SIZE]`), top index (`int top`).
- Functions: `init()`, `push(char c)`, `pop()`, and `isEmpty()`.

Drawbacks?

- Global variable management
- Functions and variables are separate
- Direct access to stack array

Stack Implementation in C

With struct

- Encapsulate stack data (but not functions)
- C structs cannot encapsulate functions

Drawbacks?

- Functions are separate from the data

Stack Implementation in C++

With struct

- Data and functions are grouped together
- C++ allows struct to include methods

Drawbacks?

- Variables are public by default

Stack Implementation in C++

With class

- Data and functions are grouped together (encapsulation)
- Promotes data hiding and abstraction